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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 31: Sustainable Transportation Planning & Approaches-II: The Contemporary Planning Process

Prof. Bhola Ram Gurjar

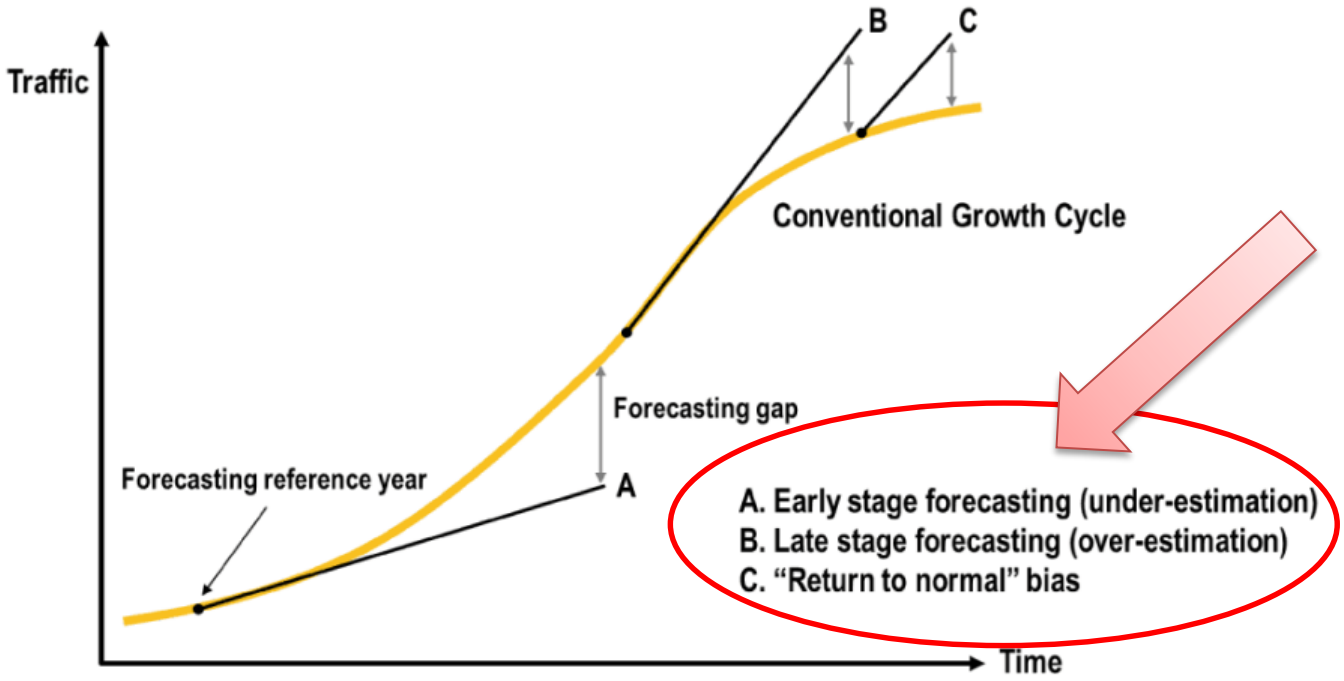
Dept. of Civil Engineering



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- Common Flaws in “Induced Planning” based on Forecasting
- Sustainable Urban Transport Planning
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- Achieving Sustainable Urban Transport Planning
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Common Flaws in “Induced Planning” based on Forecasting



Common Flaws in Planning based only on Forecasting and expansion of facilities

Source: (Jean-Paul Rodrigue 2020)

Main Challenges for Sustainable Transport Planning

- Non-inclusive development
- Difficult to meet growing demand
  - Economic and population growth, Rising income, faster than growth of economies.
- Air Pollution and GHG emissions growing faster than GDP.
- Adverse health and social impacts
- Road safety

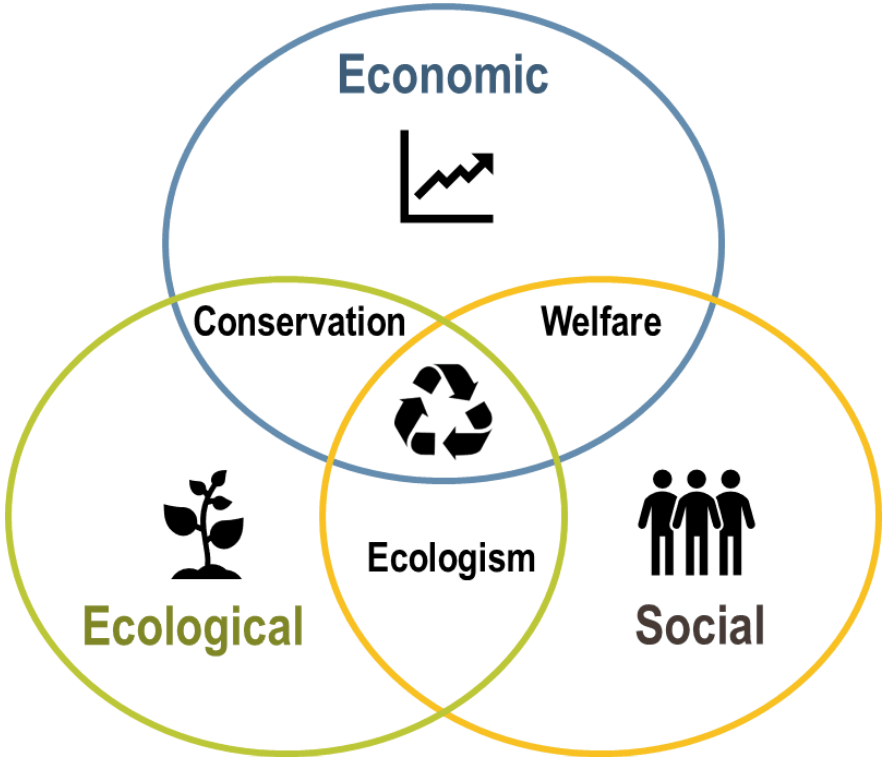
Source: (UNESCAP, 2015)

Transport Planning: View of the U.K. Transport Planning Society

- The Transport planning should be able to:
  - Prepare, assess and implement the policies, plans and projects.
  - Improve and manage the transport systems efficiently.
  - Plan on a local, regional and national level.
  - Understand the interrelationship between transport infrastructure and the future shape of cities and towns.
  - Change people's perspectives and attitude towards transportation and to encourage use of alternative modes of travel, rather than private cars.

Source: (Transport Planning Society, 2006)

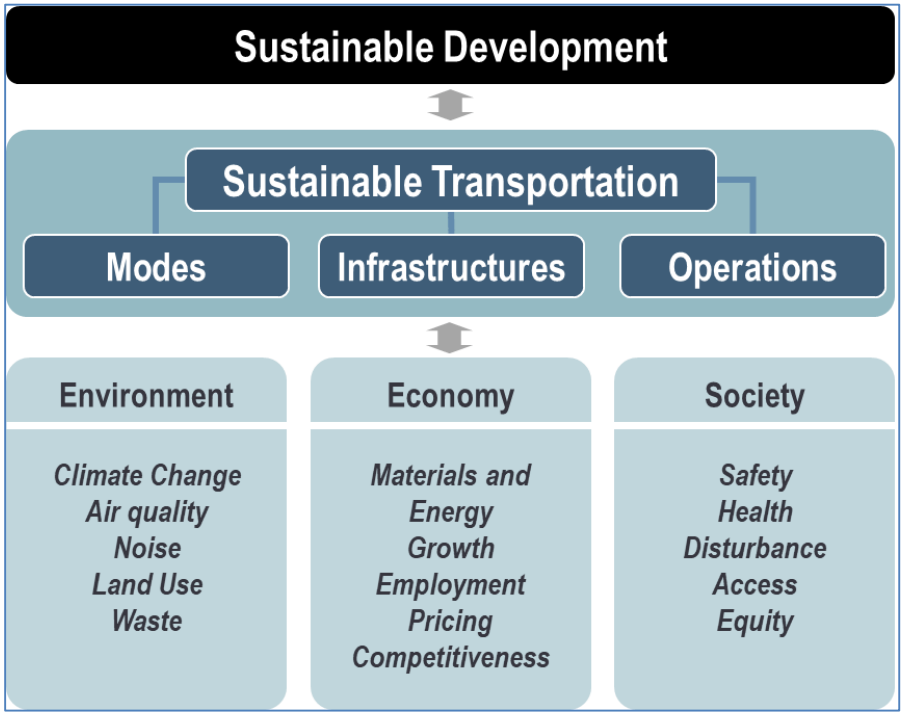
Sustainable Urban Transport Planning



Sustainable transportation is the capacity to support the mobility needs of a society in a manner that is the least damageable to the environment and does not impair the mobility needs of future generations.

Source: (Jean-Paul Rodrigue 2020)

Dimensions of Sustainable Transportation



The concept of sustainable transportation is intricately linked with the development of sustainable transport modes, infrastructures and operations.

Source: (Jean-Paul Rodrigue 2020)

The Contemporary Transport Planning Process

- Commonly Scale-specific and multi-dimensional.
- Traffic problems increased significantly since the 1970s in spite of a great deal of urban planning.
- Early Transport planning were based on Induced demand (to enhance capacities) rather than better management and maintenance of existing transport systems.



Planning Preference to Accessibility rather than Mobility

Source: (Jean-Paul Rodrigue 2020)

The Multi-stage process of Contemporary Planning

Contemporary Planning

1. Goals & Objectives

- Improving accessibility should be considered with other desirable goals such as
  - improving safety and health
  - reducing vehicular emissions
  - enhancing economic opportunities
  - improving livability
  - promoting mobility by improving equity.

2. A set of Possible options in Planning

- What objectives are desirable and what they imply must be considered in planning.
- Several alternate scenarios must be considered in the planning process.

Source: (Jean-Paul Rodrigue 2020)

The Multi-stage process of Contemporary Planning (cont'd..)



3. Identification of actors, institutions and stakeholders

- Identification of those affected and possible resolutions
- Public participation in planning

4. Predicting Outcomes, identifying benefits and assessing costs

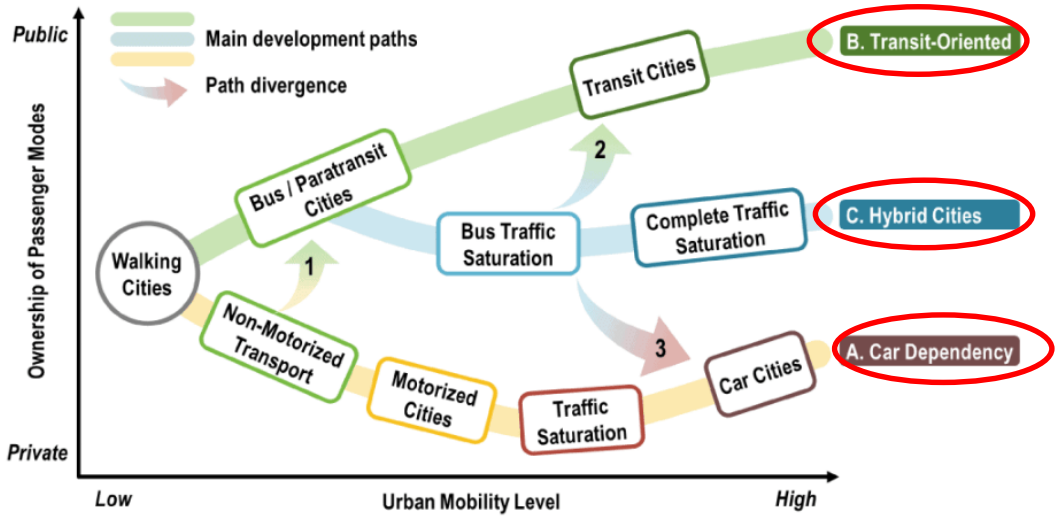
- Critical step in Planning
- Predicted outcomes and benefits must be evaluated with costs
  - Ex. Visual impacts, environmental dislocations and employment impacts

5. Choosing a course of action

- Evaluating scenarios considering costs and benefits
- Political decisions

Source: (Jean-Paul Rodrigue 2020)

Urban Transport Development Paths

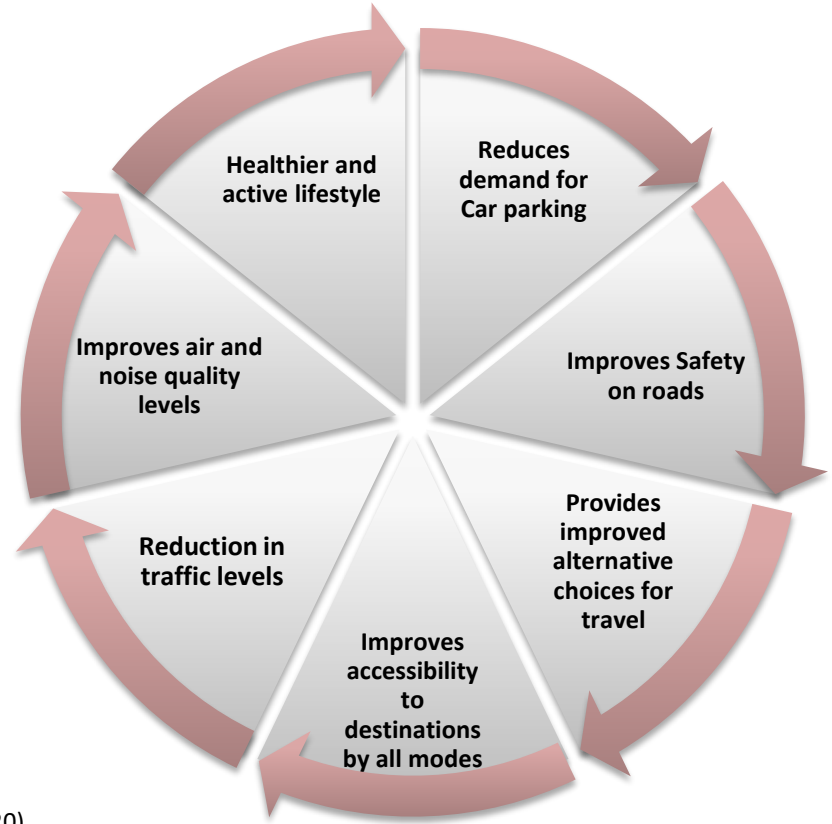


The transport development of cities is often taking place within a **path dependency**

- Results in different levels of mobility and ownership of passenger transport modes such as cars, buses, and rail transit.
- This implies that once a specific path has been elected, future developments are locked in along that path.

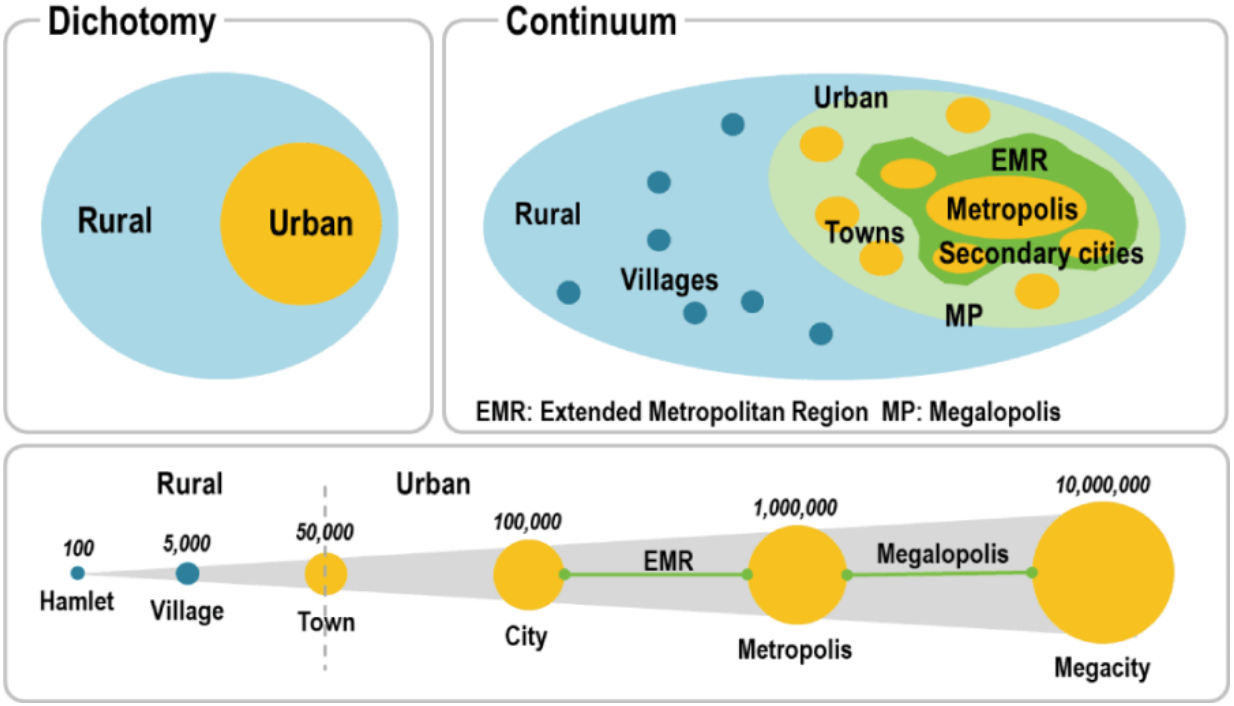
Source: (Jean-Paul Rodrigue 2020)

Benefits of reducing Car dependency



Source: (Jean-Paul Rodrigue 2020)

Planning: Dichotomy to Continuum



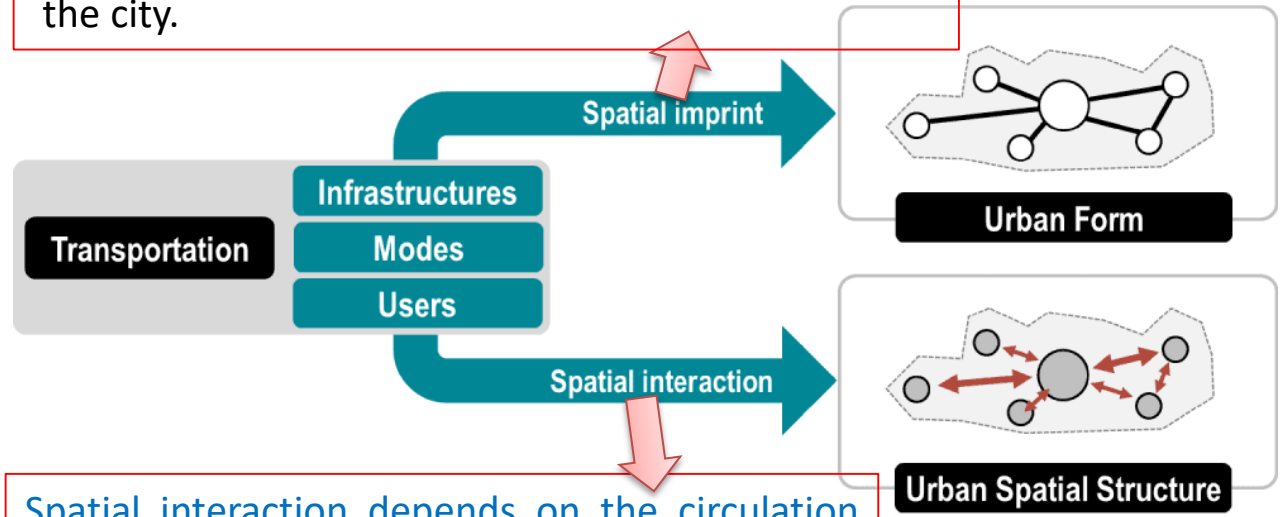
Megalopolises (mega-urban regions) are massive urban conurbations often encompassing several EMRs structured along a corridor.

Ex. BostWash (Boston – Washington), Tokaido (Tokyo – Osaka), The Pearl River Delta (Hong Kong – Guangzhou) or Shanghai – Nanjing.

Source: (Jean-Paul Rodrigue 2020)

Planning: Shift from Dichotomy to Continuum

Spatial imprint of transportation varies w.r.t. the socio-economic and geographical characteristics of the city.



Spatial interaction depends on the circulation pattern of passengers and freight inside the city.

Example.

North American cities have an urban form shaped by the automobile with dominance in highways.

Source: (Jean-Paul Rodrigue 2020)

Example of City Connectivity from the CBD

Production and Distribution

Port districts.
Heavy industries.
Railyards.
Manufacturing clusters.
Distribution clusters.

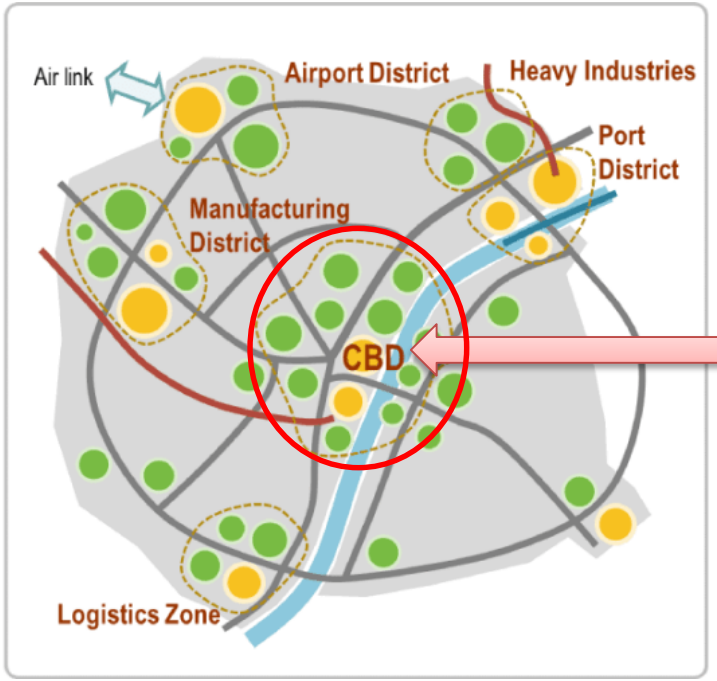
Mobility and Accessibility

Central / transit stations.
Shopping districts.
Airport districts.

Transactions

Financial / management districts.

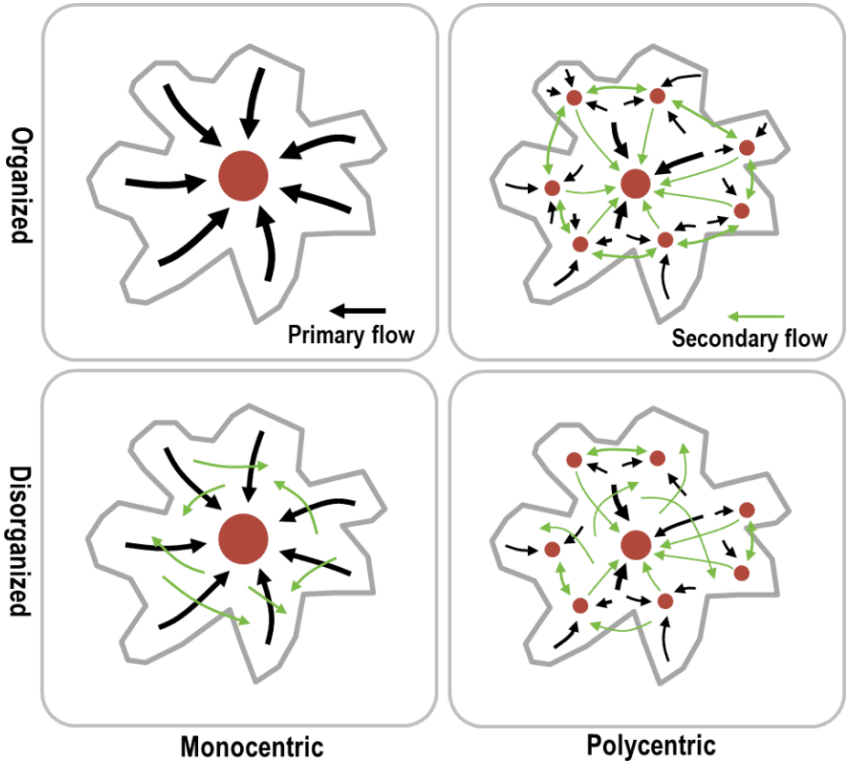
Built area
Connectivity node
Economic node
Road / transit link
Rail link
Maritime link



CBD- Central Business District

Source: (Jean-Paul Rodrigue 2020)

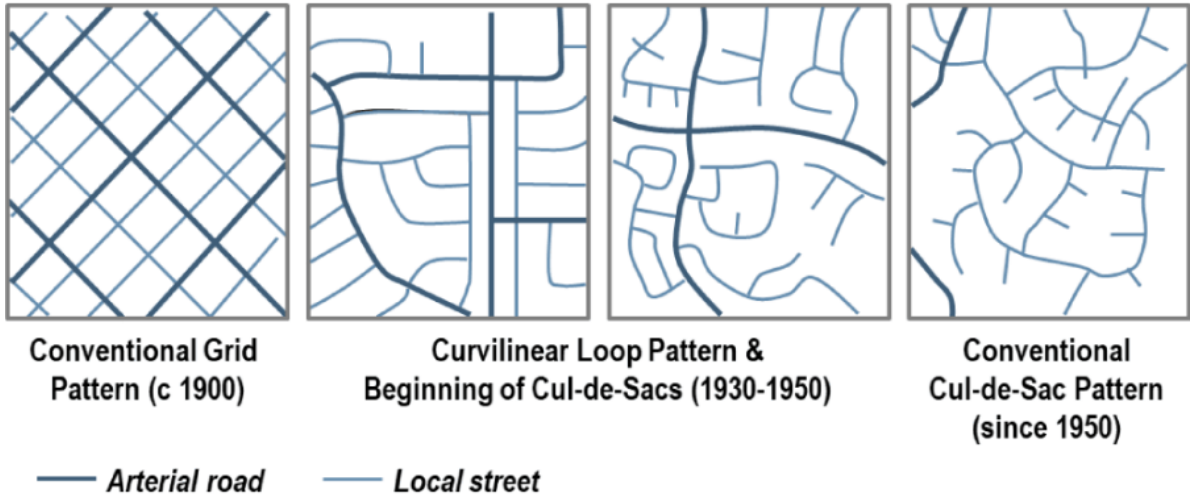
Planning Traffic flow pattern in a city



- Cities may be Polycentric (more common) or Monocentric.
- Traffic flows in a city may be Organized or disorganized (more common).

Source: (Jean-Paul Rodrigue 2020)

Planning Different Types of Street Networks



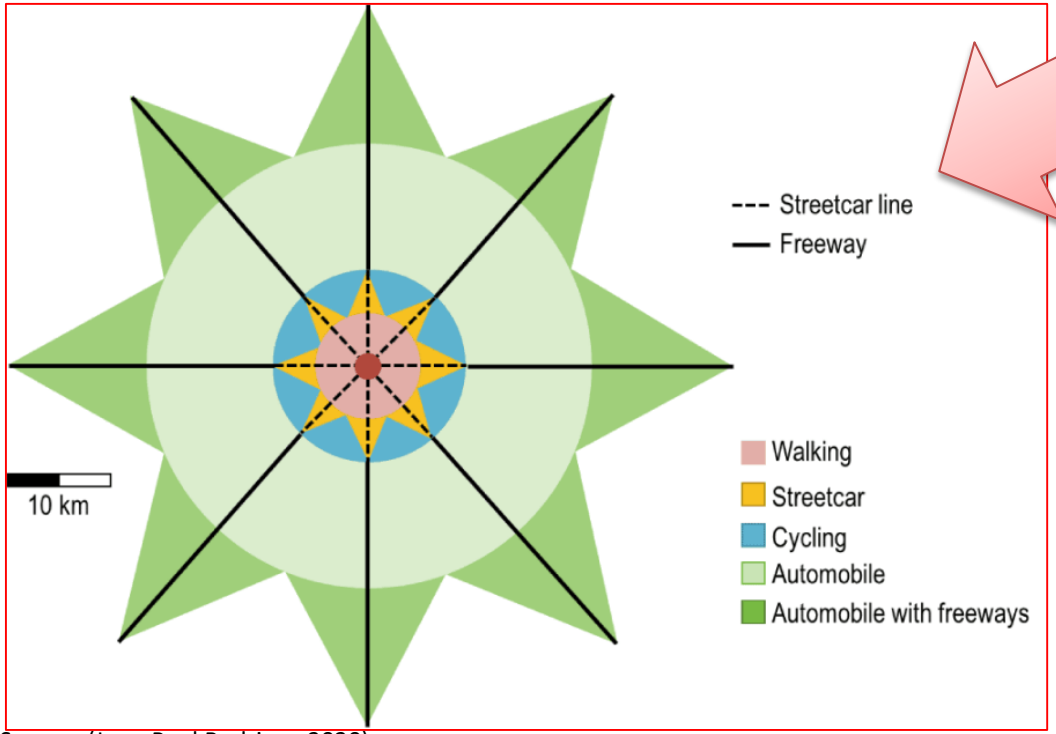
The Grid pattern was relatively not common in the 19th and Early 20th Century.

- Grid pattern: Reduction in the level of connectivity and level of landuse.
- Cul-de-sac planning to reduce and even eliminate the through movements.

Cul-de sac are dead-end streets, which reduces the extra traffic movement on a particular street mainly through residential area.

Source: (Jean-Paul Rodrigue 2020)

Example of One-hour commuting in a city for different Transport modes



Source: (Jean-Paul Rodrigue 2020)

A theoretical urban form which is the **summation of successive transport influences**.

- It assumes uniform friction of distance and capacity of transport infrastructure

Summary of Planning Transitions

S.No.	From	To
1.	Capital Planning	Management and Operations Oriented
2.	Long Timeframes	Real-time control
3.	Urban Scale Planning and Operations	Regional Scale Planning and operations
4.	Emphasis on Mobility	Emphasis on Accessibility
5.	“One size fits for All”	Customer Orientation Quality Pricing for Services
6.	Allocate Capacity by Queueing	Allocate Capacity by Pricing

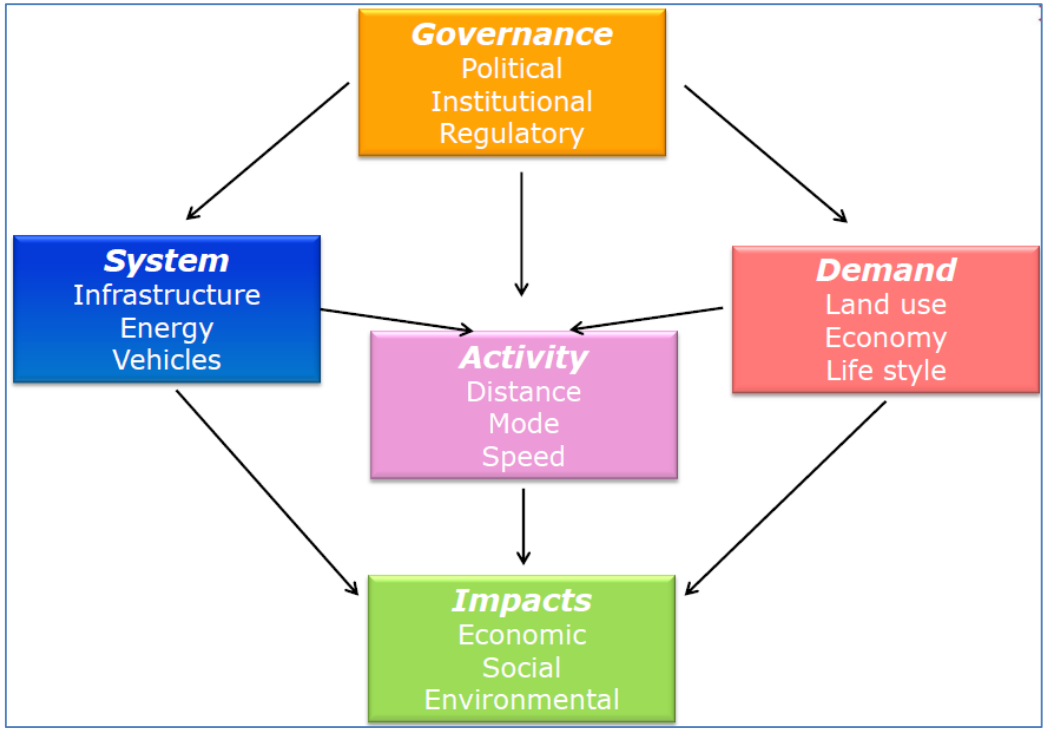
Source: (MIT lecture Notes)

Summary of Planning Transitions (cont'd..)

S.No.	From	To
7.	Aggregate methods for Demand Prediction	Disaggregate methods for Demand Prediction
8.	Public Financing	Private and Public/ Private partnerships in financing
9.	Static organizations for Institutional Relationships	Dynamic organizations for Institutional Relationships
10.	Emphasis on Design of Physical Infrastructures	Emphasis on Transportation as a complex large-scale Integrated Open system

Source: (MIT lecture Notes)

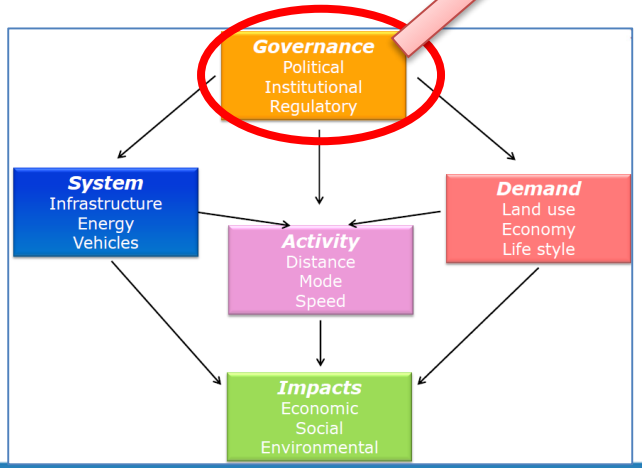
Achieving Sustainable Urban Transport Planning

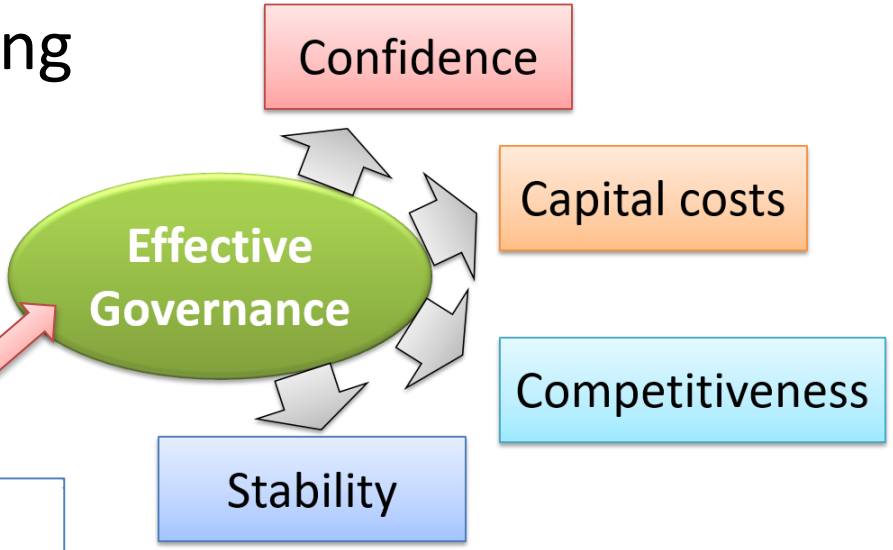


Source: (Henrik Gudmundsson, 2014)

Governance in Transport Planning

Governance of transport infrastructure is critical because of the strategic, economic and social importance of transportation

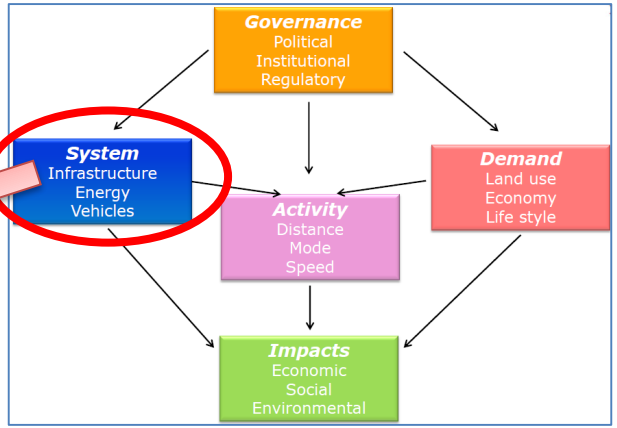


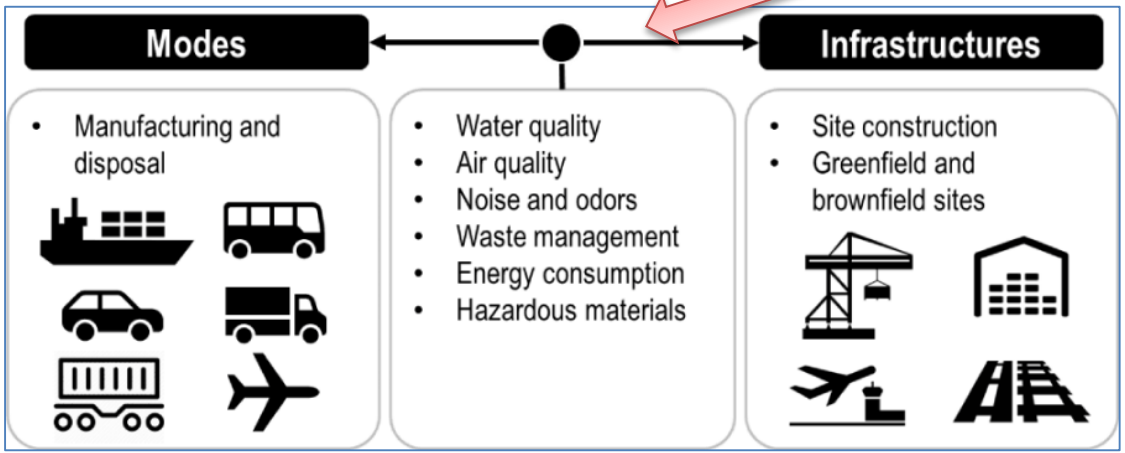


Source: (Jean-Paul Rodrigue 2020)

Modes and Infrastructure in Transport Planning

The interrelationship between the Transport modes and Infrastructures should be considered in Planning





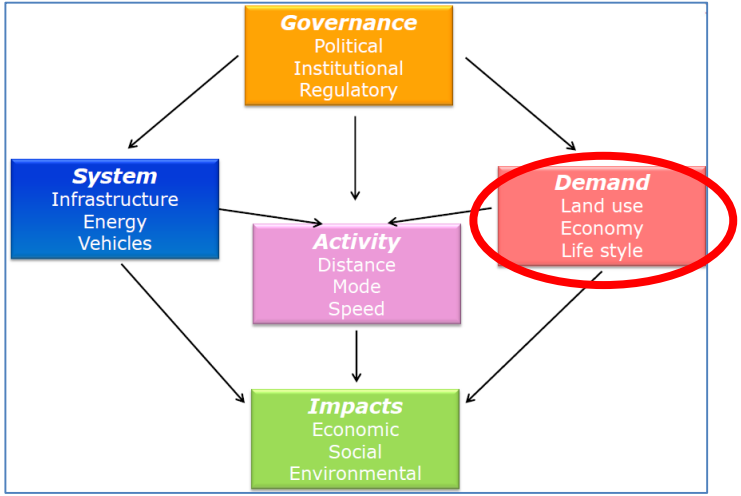
Source: (Jean-Paul Rodrigue 2020)

Transport Demand Management

- Managing both Demand and the Transport system, rather than focusing only on demand as in earlier planning processes.

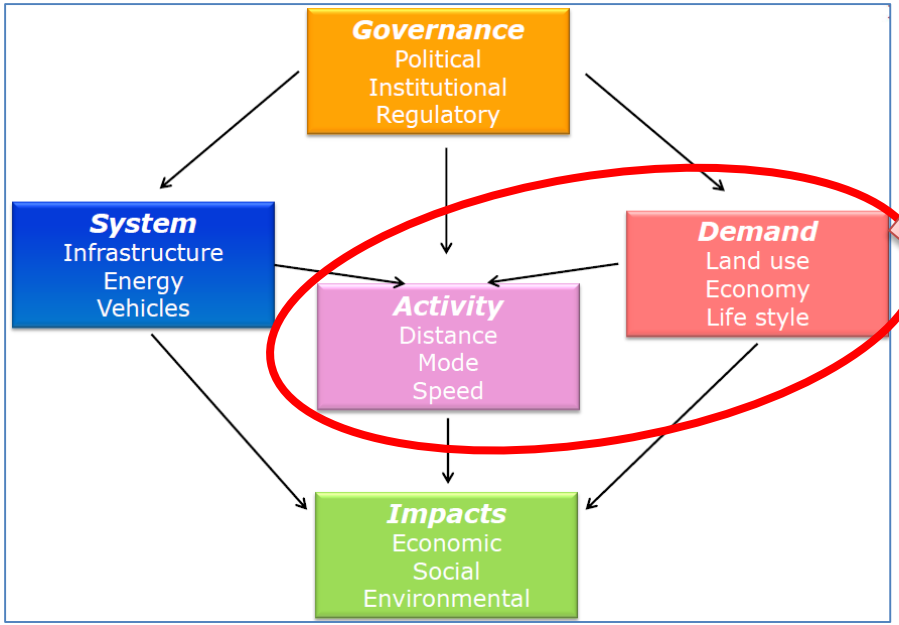
- High population densities favour walking, cycling and public transport use.

- Densification and Integration:
  - Planning concentrated along well-served transport corridors (Transit-oriented development or TOD) and increasing densities in areas undergoing rehabilitation.



Source: (Jean-Paul Rodrigue 2020)

Transport Demand Management in Planning



- Through Traffic calming measures
- Through Traffic Management strategies

Source: (Henrik Gudmundsson, 2014)

Traffic calming Measures

- Main idea is to reduce speed of flowing traffic or to reduce cut-through traffic by obstructing traffic movements in one or more directions.

- Achieved by adopting four types of measures in general:
  - Vertical deflections
  - Horizontal shifts
  - Narrowing roadways
  - Closures

Intended to reduce speed

Intended to reduce cut-through traffic by obstructing traffic movements in one or more directions

Speed Humps

- Vertical obstructions, rounded and raised areas of pavement, also known as road humps or undulations on roads.

- Used to slow cars to a speed of about 30km/h or lower in their direct vicinity.

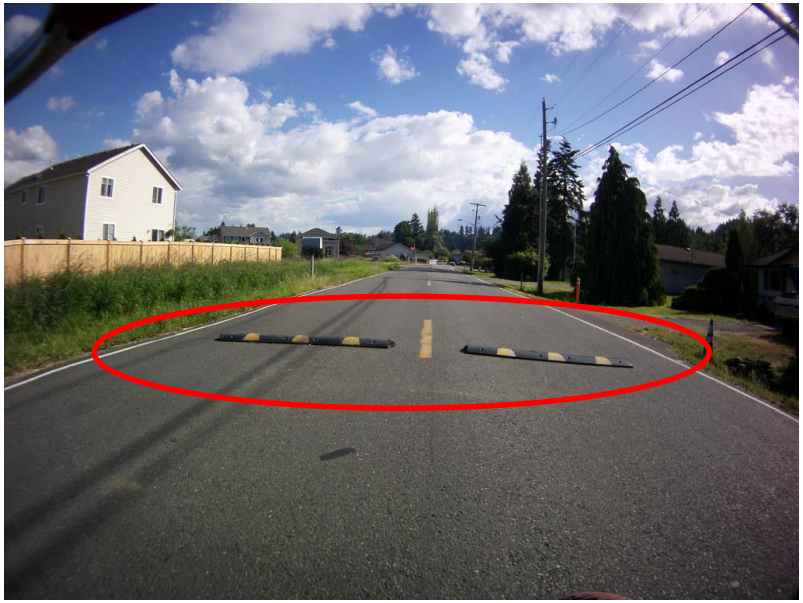
- Not typically used on major roads, bus routes, or primary emergency response routes.



Source: (Institute of Transportation Engineers (ITE), https://www.ite.org/technical-resources/traffic-calming/traffic-calming-measures/)

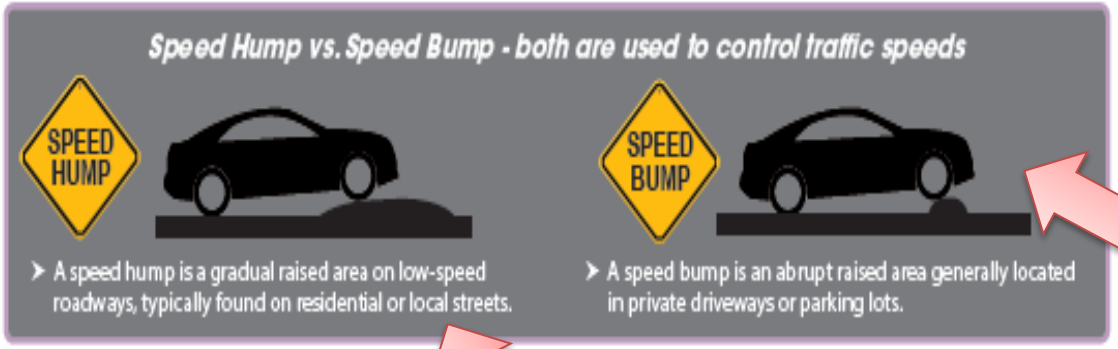
Speed Bumps

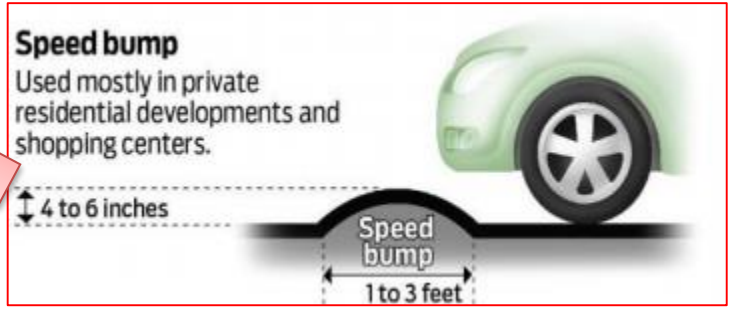
- Similar to Speed humps, but more aggressive in function.
- Forces vehicles to slow down much slower than that of speed humps.
- Used in parking lots, residential streets etc.

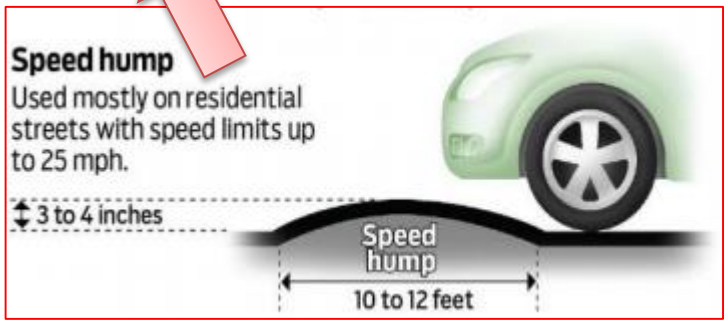


Source: (Institute of Transportation Engineers (ITE), https://www.ite.org/technical-resources/traffic-calming/traffic-calming-measures/)

Speed Humps vs Speed Bumps







Source: (https://safety.fhwa.dot.gov/speedmgt/ref_mats/fhwasa16077/)

Speed Table

- Speed Tables are similar to speed humps, but are long, raised humps with a flat section in the middle and ramps on the ends.

- Referred to as a raised crosswalk, if placed at a pedestrian crossing.

- Referred to as an Offset speed table, if placed only at one direction of a road.



Source: (https://trafficlogix.com/speed-tables/)

Source: (Institute of Transportation Engineers (ITE), https://www.ite.org/technical-resources/traffic-calming/traffic-calming-measures/)

Curb Extensions/ Bulb-Out

- Horizontal extension of the sidewalk into the street, resulting in a narrower roadway section.
- Curb extensions make it easier and safer for pedestrians to cross the street by shortening the distance from curb to curb.
- It is called a choker, if placed at a mid-block location.



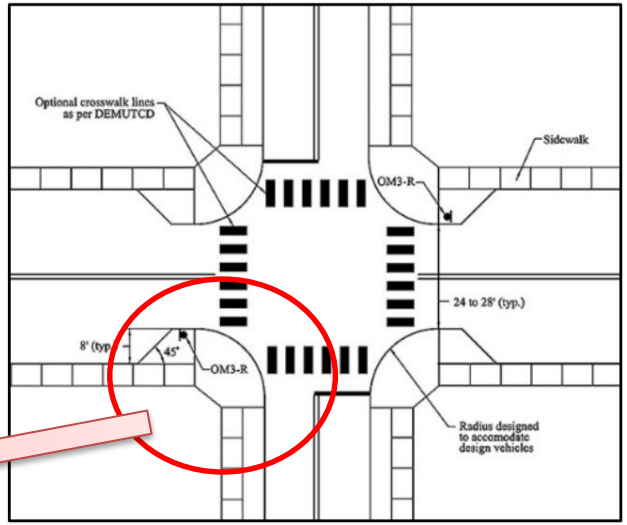
Source: (Institute of Transportation Engineers (ITE), https://www.ite.org/technical-resources/traffic-calming/traffic-calming-measures/)

Example of Curb Extensions

Sidewalk extensions provide safety to pedestrians while providing space for benches and street lighting.



Source: (https://www.reliance-foundry.com/blog/traffic-calming-bollards)



(Source: Delaware DOT)

Other examples of Curb Extensions/ Bulb-Out at Intersections

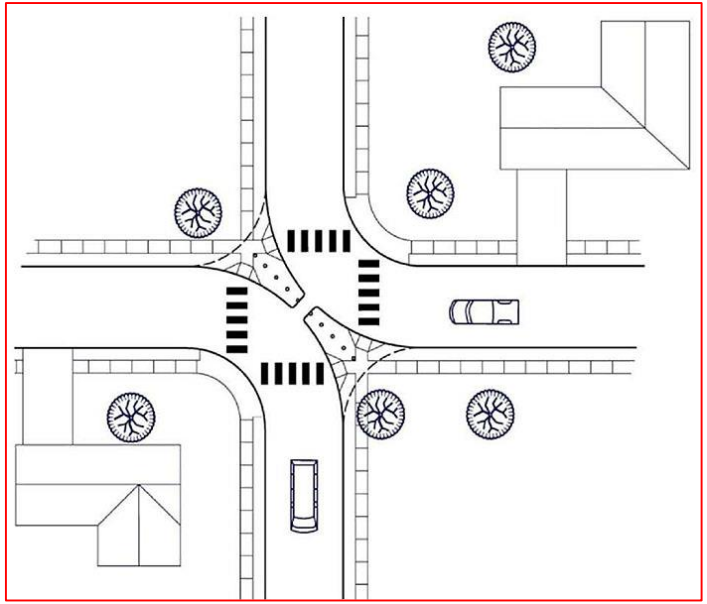




Source: (https://abaat.org/traffic/curb-extensions/)

Diagonal Diverter

- These are barriers placed diagonally across four-legged intersections, blocking through movements.
- Also known as Full diverters or diagonal road closures.
- Sometimes, diagonal diverters can be used as partial road closures, allowing movement of only two-wheelers and obstructing movements of four-wheelers.



Source: (https://safety.fhwa.dot.gov/speedmgt/ePrimer_modules/module3pt3.cfm)

Source: (Institute of Transportation Engineers (ITE), https://www.ite.org/technical-resources/traffic-calming/traffic-calming-measures/)

Example of Full Road closure using Diagonal Diverter



Source: (https://safety.fhwa.dot.gov/speedmgt/ePrimer_modules/module3pt3.cfm)

Example of Partial Road closure using Diagonal Diverter



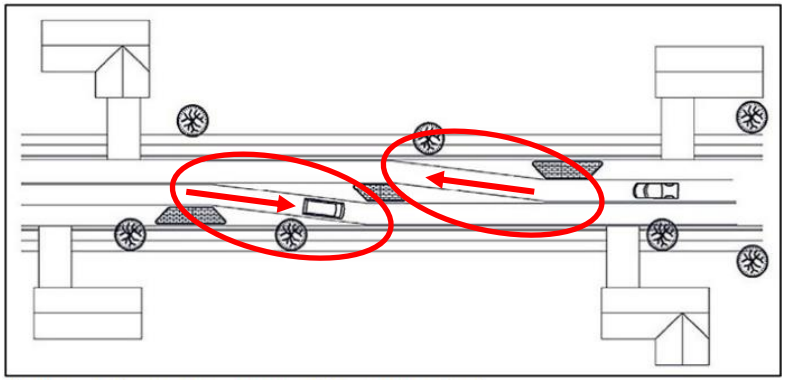


Source: (https://safety.fhwa.dot.gov/speedmgt/ePrimer_modules/module3pt3.cfm)

Lateral Shift

- Realignment of an otherwise straight street that causes travel lanes to shift in at least one direction.

- Applications:
  - Appropriate for local, collector, or arterial roadways
  - Appropriate for one-lane one-way and two-lane two-way streets
  - Appropriate on roads with or without dedicated bicycle facilities
  - Maximum appropriate speed limit is typically 35 mph.
  - Appropriate along bus transit routes.



(Source: Delaware Department of Transportation)

Source: (Institute of Transportation Engineers (ITE), https://www.ite.org/technical-resources/traffic-calming/traffic-calming-measures/)

Lateral Shift

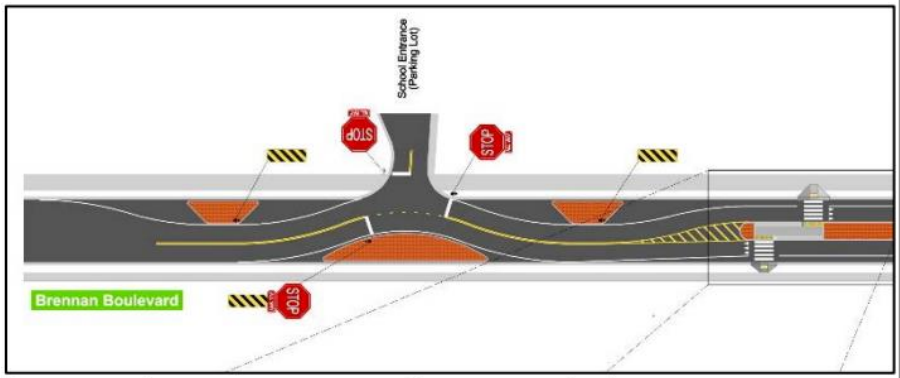




Source: (https://www.fhwa.dot.gov/publications/research/safety/15030/009.cfm)

Chicanes

- A series of alternating curves or lane shifts that force a motorist to steer back and forth instead of traveling a straight path and creates extra turns along a road to slow traffic.
- Applications:
  - Appropriate for mid-block locations but can be an entire block if it is relatively short
  - Appropriate speed limit is typically 35 mph or less.
  - Applicable on one-lane one-way and two-lane two-way roadways
  - Can be used with either open or closed (i.e. curb and gutter) cross-section



(Source: Delaware Department of Transportation)

Source: (Institute of Transportation Engineers (ITE), https://www.ite.org/technical-resources/traffic-calming/traffic-calming-measures/)

Chicanes



Chicanes force vehicles to slow down and zig-zag around the obstacles.

Source: (https://www.reliance-foundry.com/blog/traffic-calming-bollards)

Road Closures



- May be Half closures or Full closures.

- Full-street closures are barriers placed across a street to completely close the street to through traffic, usually leaving open space for pedestrians and bicyclists; they are sometimes called cul-de-sacs, dead-ends, or mini-parks.

- Half closures are barriers that block travel in one direction (creates a one-way street) for a short distance on otherwise two-way streets; sometimes called partial closures or one-way closures.

Source: (Institute of Transportation Engineers (ITE), https://www.ite.org/technical-resources/traffic-calming/traffic-calming-measures/)

Example of Half Road closure





Source: (https://safety.fhwa.dot.gov/speedmgt/ePrimer_modules/module3pt3.cfm)

Example of Full Road closure



Source: (http://www.stocktongov.com/government/departments/publicWorks/tCalmChoiceFull.html)

Median Barrier/ Forced Turn Island

- Raised islands along the centreline of a street and continuing through an intersection

- Blocks the through movement from the cross street, also called median diverter, intersection barrier, intersection diverter, and island diverter.



Source: (https://safety.fhwa.dot.gov/speedmgt/ePrimer_modules/module3pt3.cfm)

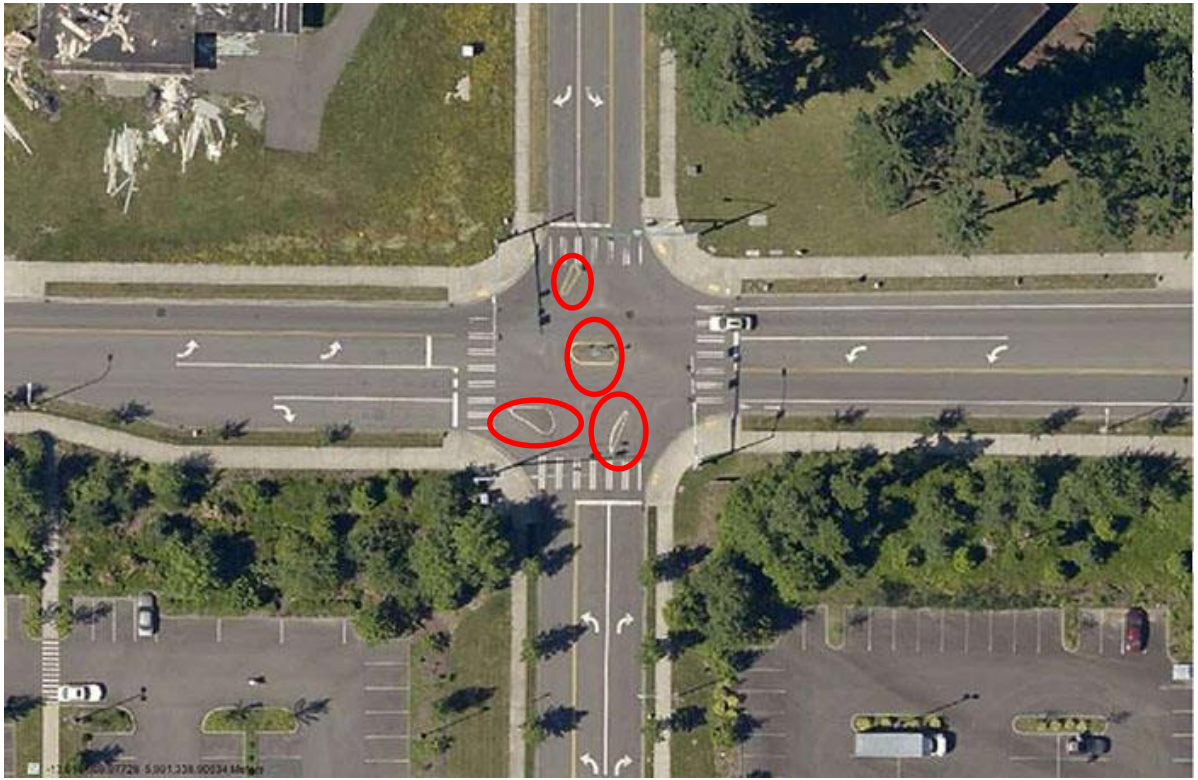
Source: (Institute of Transportation Engineers (ITE), https://www.ite.org/technical-resources/traffic-calming/traffic-calming-measures/)

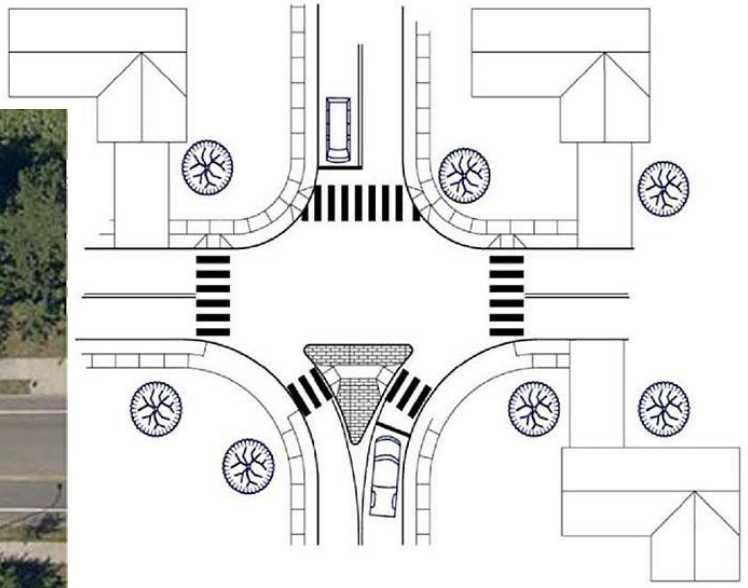
Example of Median Barrier with Pedestrian Refuge and Bicycle Cut-Through



Source: (https://safety.fhwa.dot.gov/speedmgt/ePrimer_modules/module3pt3.cfm)

Example of Forced Turn Island





Source: (https://safety.fhwa.dot.gov/speedmgt/ePrimer_modules/module3pt3.cfm)

Median Islands

- Raised island located along the street centreline that narrows the travel lanes at that location.
- Also called median diverter, intersection barrier, intersection diverter, and island diverter.
- Median islands separate opposing directions of traffic, and can slow traffic by briefly narrowing the roadway.



Source: (https://www.reliance-foundry.com/blog/traffic-calming-bollards)

Source: (Institute of Transportation Engineers (ITE), https://www.ite.org/technical-resources/traffic-calming/traffic-calming-measures/)

Median Islands





Median Island with cross walk

Source: (https://safety.fhwa.dot.gov/speedmgt/ePrimer_modules/module3pt3.cfm)

Mini Roundabout

- Raised islands, placed in unsignalized intersections, around which traffic circulates.

- Require drivers to slow to a speed that allows them to comfortably maneuver around them.



Source: (Institute of Transportation Engineers (ITE), https://www.ite.org/technical-resources/traffic-calming/traffic-calming-measures/)

Examples of Mini Roundabouts



Source: (https://www.msa-ps.com/compact-and-mini-roundabouts-maximized-safety-at-minimum-size/)



Source: (nacto.org)

Roundabout

- Raised islands which are large in diameter and placed in unsignalized intersections around which traffic circulates
- Requires drivers to slow to a speed that allows them to comfortably maneuver around them.
- Can be used at intersections with high volumes of large trucks and buses, depending on design.
- Different from traffic circles or mini-roundabouts; possible substitute for traffic signal control.



Source: (https://www.chandigarhtrafficpolice.gov.in/use-indicators-signals.php)

Source: (Institute of Transportation Engineers (ITE), https://www.ite.org/technical-resources/traffic-calming/traffic-calming-measures/)

Example of Roundabout



Roundabouts encourage steady flow of traffic and can include amenity features such as statues, fountains, and community gardens

On-Street Parking

- Allocation of paved space to parking.
- Narrows road travel lanes and increases side friction to traffic flow.
- Can apply on one or both sides of roadway.
- Can be either parallel or angled, but parallel is generally preferred for maximized speed reduction.



Source: (Institute of Transportation Engineers (ITE), https://www.ite.org/technical-resources/traffic-calming/traffic-calming-measures/)

Examples of On-street Diagonal Parking





Source: (https://www.pps.org/article/livememtraffic)

Example of On-street Parallel Parking



Raised Pedestrian crossings or Intersections

- Flat raised areas covering entire intersections, with ramps on all approaches and often with brick or other textured materials on the flat section and ramps.

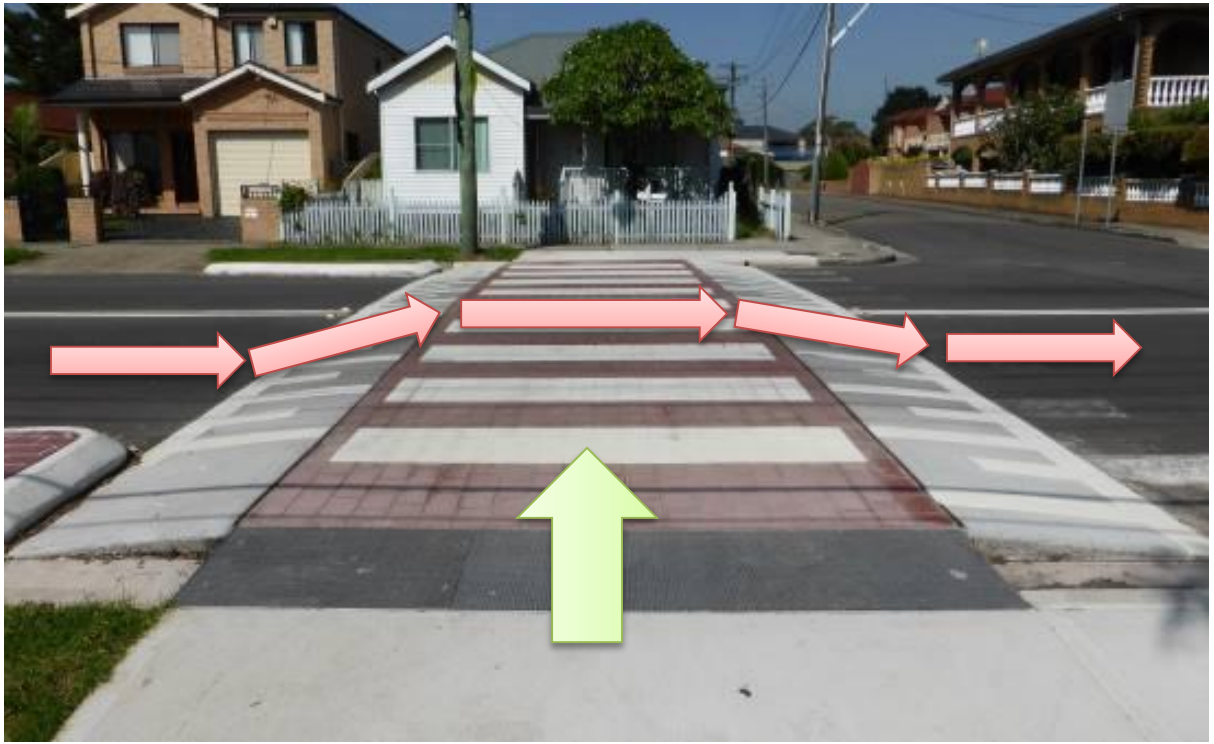
- Raising a section of the roadway to the height of the sidewalk slows vehicles and makes it easier for pedestrians to cross the street.

- Sometimes referred to as raised junctions, intersection humps, or plateaus.



Source: (Institute of Transportation Engineers (ITE), https://www.ite.org/technical-resources/traffic-calming/traffic-calming-measures/)

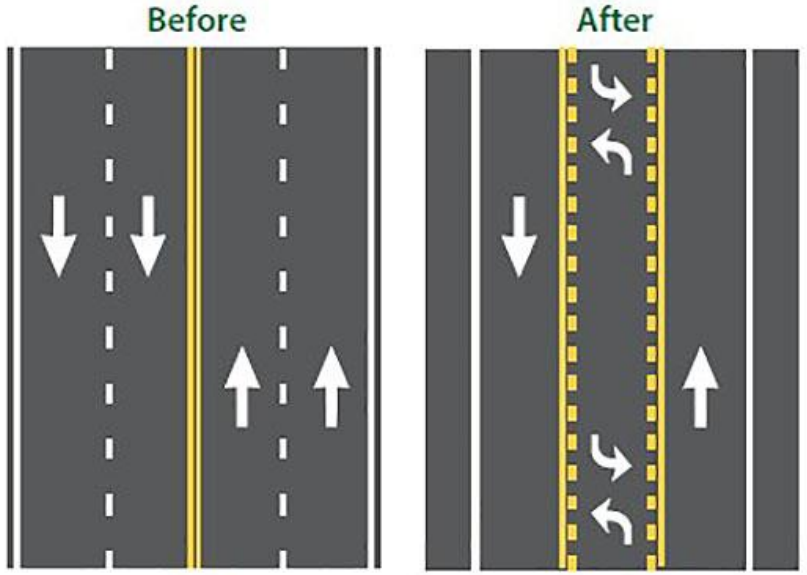
Example of Raised Pedestrian crossings



Raised Pedestrian crossings enables easy and same level crossings for all pedestrians and specially abled people, while reducing speed for vehicles

Road Diet

- Revision of lane use or widths to result in one travel lane per direction with minimum practical width, with goal of reducing cross-section.
  - Example: Conversion of four-lane Two-way road to three-lane road – two through lanes and center two-way left-turn lane (TWLTL).
- Can also involve narrowing of existing travel lanes
- Alternate cross-section uses can include dedicated bicycle facilities, left-turn lanes, on-street parking, raised medians, pedestrian refuge islands, sidewalks, etc.



(Source: FHWA Road Diet Information Guide)

Source: (Institute of Transportation Engineers (ITE), https://www.ite.org/technical-resources/traffic-calming/traffic-calming-measures/)

Example of Before and After Road Diet





In this example, the Road width is reduced and converted as On-street parking areas

Source: (https://safety.fhwa.dot.gov/speedmgt/ePrimer_modules/module3pt3.cfm)

Example of Road Diet with Bicycle lanes



In this example, the Road width is reduced and converted as Bicycle lanes

Source: (https://safety.fhwa.dot.gov/speedmgt/ePrimer_modules/module3pt3.cfm)

Example of Before and After Road Diet



Example of Seattle's Rainier Avenue Street's 4-mile dangerous lane with people driving at 40 mph+, was modified after which the speeds got decreased by approx. 95%.

Source: (https://seattlegreenways.org/blog/tag/road-diet/)

Conclusion

- The Contemporary Transport planning ensures Sustainable development by emphasising on Accessibility rather than Mobility.

- Transport planning should be focused more on the management of existing traffic and infrastructure rather than mere capacity enhancements.

- Traffic calming measures ensures safe and focused planning through controlling traffic speed and volume in the city.

- Traffic management strategies should also be combined with calming measures and policy measures for effective results.

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Thank You